

**SIMATS ENGINEERING
ASSESSMENT TOOL**

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:20

Scenario Based

Code of Conduct:

I V. Smylika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Smylika.✓
Signature of the student:

CO3 Assessment Task 3DFA Minimization for PIN Validation

The DFA validates PIN patterns such as bbbe, bco, co, caaaaa, etc.

Minimization

To minimize the DFA:

1. Remove unreachable states.
2. Separate final and non-final states.
3. Compare transitions of the remaining states.
4. Merge equivalent states.

The resulting DFA has fewer states while accepting the same PIN patterns.

Advantages:

- Reduces memory usage.
- Requires fewer logic gates.
- Faster PIN validation.
- Reduces power consumption.
- Suitable for resource-limited embedded systems.

Conclusion:

DFA minimization improves both speed and hardware efficiency without changing the accepted PIN patterns.

2. Finite Automaton to Regular Expression
The given finite automaton can be converted into a regular expression using state elimination.

From the diagram:

- q_1 is the initial state.
- q_4 is the final state.
- q_1 has a loop on 0.
- $q_1 \rightarrow q_2$ on 0.
- $q_2 \rightarrow q_1$ on 1.
- $q_2 \rightarrow q_3$ on 1.
- $q_3 \rightarrow q_1$ on 0.
- $q_3 \rightarrow q_4$ on 0.
- q_4 has a loop on 0, 1.

A suitable expression is:

$$0^* (00^*1 + 10)^* (00 + 11)(0 + 1)^*$$

3. Grammar:

Given:

$$S \rightarrow aSc \mid T \mid U \mid \epsilon$$

$$T \rightarrow bTc \mid \epsilon$$

$$U \rightarrow aUbc \mid \epsilon$$

A) Language

The grammar generates:

$$L = \{a^n b^n c^n\} \cup \{b^n c^n\} \cup \{a^n c^n\}, n \geq 0$$

B) Derivation of aaabbc

$$S \Rightarrow aSc$$

$$\Rightarrow aaUbc$$

$$\Rightarrow aaaUbbc$$

$$\Rightarrow aaabbc$$

$$\therefore S \Rightarrow^+ aaabbc$$

C) Ambiguity

The grammar is ambiguous if a word has two different derivations.

For this grammar, the alternatives γ and U generate different forms, so the same non-empty word cannot normally be generated through both.

4. Regular Languages and Prime Language

i) Prove

$L_1 \cup L_2$ is regular ✓

IF

L_1

and

L_2 are regular, there exist finite automata for both.

A new automaton can combine them using ϵ -transitions from start state to the start states of the two automata.

$\therefore L_1 \cup L_2$ is regular.

Hence, regular languages are closed under union.

ii) Is

$a^p \mid p$ is prime regular?

No, it is not regular.

A finite automaton has only a finite number of states and cannot determine whether an arbitrarily large number of symbols is prime.

Therefore:

$\{p \mid p \text{ is prime}\}$ is not regular.

SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

89/100